

OIML R-76 METROLOGY LABORATORY

NON-AUTOMATIC WEIGHING INSTRUMENT (NAWI) VERIFICATION

Report Ref: R76-2026-TEST01
Cert No: CERT-2026-8889
Date: 2026-09-07
Test Type: Initial Verification (Type Approval/New)

OVERALL METROLOGICAL
RESULT: PASS

Standard: OIML R-76-1:2006 (E)
Turning Point Method: Enabled (Small Weights)
Instrument meets all tested requirements of OIML R-76-1.

1. NAWI & METROLOGICAL SPECIFICATIONS

Manufacturer:	Avery Weigh-Tronix			Accuracy Class:	Class III (Medium Accuracy)		
Model / Type:	ZK830-15K			Max Capacity (Max):	15.0 kg		
Serial Number:	AW-2026-99124			Min Capacity (Min):	0.1 kg		
Device Type:	Retail Counter NAWI Scale			Scale Interval (e):	0.005 kg		
Customer / Location:	Apex Food Logistics Ltd / Central Testing Lab - Bay 4			Actual Interval (d):	0.005 kg		
Inspector / Verifier:	R. Sharma (Metrology Eng.)			Divisions n = Max/e:	3,000 divisions		

Temperature:	21.5 °C	Humidity:	48.0 % RH	Pressure:	1012.0 hPa	Weights Ref:	OIML Class M1 Standard Weights Set (1g - 20kg) (NABL-CAL-2026-00441)
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2. SUMMARY OF METROLOGICAL EVALUATIONS (OIML R-76-1)

OIML R-76 Clause / Evaluation Subject	Requirement / Criterion	Result
Instrument Metrological Limits (Table 3)	divisions n within class range; Min capacity verified	PASS
Zero Setting Accuracy (A.4.2)	Zero error E0 <= 0.25 e	PASS
Tare Balancing & Net Weighing (A.4.6)	Net indication corrected error <= MPE	PASS
Weighing Performance & Hysteresis (A.4.4)	Increasing & Decreasing load errors <= MPE	PASS
Repeatability (A.4.10)	Span delta (Imax - Imin) <= MPE	PASS
Eccentricity (A.4.7)	Errors at 5 load receptor positions <= MPE	PASS

3. DETAILED METROLOGICAL TEST PROTOCOLS

Test Description	Nominal Load	Indication	Small Wt (ΔL)	Error (Ec)	MPE / Limit	Status
Zero Setting (A.4.2)	0.00 kg	0.0 kg	0.0025 kg	0.0 kg	± 0.00125 kg (0.25e)	PASS
Tare Balancing (A.4.6)	2.0 kg	2.0 kg	0.0025 kg	0.0 kg	± 0.0013 kg	PASS
Tare Net Weighing (A.4.6)	Net: 5.0 kg	5.0 kg	0.0025 kg	0.0 kg	± 0.005 kg	PASS

Eccentricity Test (clause A.4.7) — Test Load: 5.0 kg | MPE: ± 0.005 kg | Max Diff: 0.001 kg

Pos #	Position Name / Description	Indication	Small Wt (ΔL)	Error (E)	Corrected (Ec)	Status
1	Center (1)	5.0 kg	0.0025 kg	0.0 kg	0.0 kg	PASS
2	Front-Left (2)	5.0 kg	0.002 kg	0.0005 kg	0.0005 kg	PASS
3	Back-Left (3)	5.0 kg	0.003 kg	-0.0005 kg	-0.0005 kg	PASS
4	Back-Right (4)	5.0 kg	0.0025 kg	0.0 kg	0.0 kg	PASS
5	Front-Right (5)	5.0 kg	0.0025 kg	0.0 kg	0.0 kg	PASS

Repeatability Test (clause A.4.10) — Requirement: Δ = (I_max - I_min) ≤ |MPE|

Load Level	Observed Indications (Runs 1..N)	I_max	I_min	Spread (Δ)	MPE	Status
7.5 kg	7.5, 7.5, 7.505	7.505 kg	7.5 kg	0.005 kg	± 0.005 kg	PASS
15.0 kg	15, 15.005, 15	15.005 kg	15.0 kg	0.005 kg	± 0.0075 kg	PASS

Weighing Performance Test (clause A.4.4) — Max Hysteresis: 0.0 kg

Load (L)	I (Inc)	Ec (Inc)	I (Dec)	Ec (Dec)	Hysteresis	MPE ($\pm$)	Status
0.1 kg	0.1 kg	0.0	0.1 kg	0.0 kg	0.0 kg	± 0.0025 kg	PASS
2.5 kg	2.5 kg	0.0	2.5 kg	0.0 kg	0.0 kg	± 0.0025 kg	PASS
5.0 kg	5.0 kg	0.0	5.0 kg	0.0 kg	0.0 kg	± 0.005 kg	PASS
10.0 kg	10.0 kg	0.0	10.0 kg	0.0 kg	0.0 kg	± 0.005 kg	PASS
15.0 kg	15.0 kg	0.0	15.0 kg	0.0 kg	0.0 kg	± 0.0075 kg	PASS

Tested / Verified By:	Approved / Authorized By:	Official Seal & Verification Date:
		Date: 2026-09-07
Technical Metrologist / Inspector	Quality Manager / Signatory	Place: Central Testing Lab - Bay 4
		Ref: R76-2026-TEST01

METROLOGICAL NOTICE & STATUTORY DISCLAIMER:
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